

Ziwei Guo

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Education

Vanderbilt University, Nashville, Tennessee

Aug 2024 – May 2026

M.S. in Computer Science, GPA: 3.75/4.00

Dickinson College, Carlisle, Pennsylvania

Aug 2020 – May 2024

B.S. in Mathematics & Data Analytics, Minor in Music

Technical Skills

Languages: Python, Go, JavaScript/TypeScript, SQL

Infra & Distributed Systems: Kubernetes, Docker, Kafka, Redis, Vault, AWS, Linux/Unix, CI/CD, Railway, Vercel

Observability: Prometheus, Grafana, PromQL, Zap Logging, metrics, logs, alerting, telemetry, Kubernetes rollouts

Backend & Data Systems: PostgreSQL, FastAPI, Flask, Node.js, Express.js, WebSockets, SSE, REST APIs, data pipelines, PyArrow, Parquet, DuckDB

AI/Agent Systems: RAG, LLM fine-tuning, ReAct, agent orchestration, MCP, Playwright, Stagehand, LLM evaluation

Frontend: React.js, Next.js, HTML/CSS, Tailwind CSS, D3.js

Experience

Tesla, Inc. — Factory Software (FSS), Fremont, CA

May 2025 – Aug 2025

Software Engineer Intern

- Built **FSS-Agent**, a production observability and diagnostics platform deployed across Fremont, Reno, and Buffalo; translated natural-language operational questions into structured Prometheus/Grafana KPI health checks over WebSocket and REST APIs, reducing incident triage time by **30%+** and saving **10+ engineer-hours per week**.
- Replaced static YAML configuration with **Grafana API-driven discovery** of PromQL queries, thresholds, and dashboard metadata from **5 production dashboards**, enabling runtime updates, structured diagnostics, and faster debugging across **3 factory sites** without code changes.
- Engineered Kafka TLS certificate monitoring and automated rotation using cert-manager, Vault, Prometheus, and Kubernetes; validated certificate metadata and secret consistency, then safely rolled **5 production services** to prevent credential-expiration outages.
- Built a Nuclio-based, multi-cluster service-account rotation system with Vault updates, database validation, deduplication, retries, delayed-rollout safeguards, rolling restarts, and HTML alerts for **12+ production services**.

Projects

AgentCert — Continuous Assurance Platform for Action-Taking AI Agents

2026

- Built and deployed a hosted Python and TypeScript platform for continuously testing, controlling, and verifying AI-agent actions; launched at **agentcert.app** and published **agentcert@0.6.3** with one-command CLI onboarding, GitHub Actions integration, and JSON/JUnit/HTML/Markdown outputs.
- Developed three assurance engines: **MCPBench** for runtime MCP/tool-use evaluation and sensitive-data-flow checks, **Tripwire CI** for browser-agent fault and prompt-injection testing, and **Onegent Runtime** for policy-based **ALLOW / REQUIRE APPROVAL / BLOCK** decisions, human approvals, and post-action outcome verification.
- Implemented continuous re-assurance across PR, release, and nightly tiers using scope fingerprinting and lifecycle states (**CURRENT / REVALIDATION REQUIRED / SUSPENDED / EXPIRED**), with hash-linked signed records, server attestation, and SHA-256 artifact validation.
- Delivered a multi-tenant PostgreSQL backend with RBAC, API keys, idempotency, replay protection, incidents, webhook alerts, policy and decision history, and reproducible audit evidence backed by **50+ automated tests**.

Market Data Integrity & Replay Platform — Nasdaq ITCH 5.0 / MoldUDP64

2026

- Built an official-spec Python market-data system supporting all **23 Nasdaq TotalView-ITCH 5.0 message types** and MoldUDP64 sequencing, including strict binary validation, exact gap detection, duplicate suppression, A/B feed arbitration, order-book reconstruction, and deterministic archive recovery.
- Scanned a **12.95 GB full-day feed containing 423,285,709 messages** and completed repeatable AAPL/MSFT replays in bounded memory with zero parse failures, unknown records, or state-invariant violations.
- Produced point-in-time feature, trade, and order-lifecycle tables through incremental PyArrow/Parquet writers, with DuckDB querying, event-level lineage, checkpoint hashes, and logical outputs independent of safe batch boundaries.
- Verified every supported message against an independent audit parser and **itchfeed 1.0.7** with zero differences; executed **1,000 deterministic fault scenarios** and achieved a **2.33× replay speedup** without changing results across **215 passing tests** and **93.44% coverage**.

GridQueue Agent — Public Grid Intelligence and High-Stakes Agent Evaluation

2026

- Built and deployed a FastAPI, DuckDB, and Next.js pipeline that ingests public interconnection-queue files, normalizes records, resolves entities across monthly snapshots, detects material changes, and generates source-grounded risk briefs with citations, caveats, and reproducibility traces.
- Processed real ERCOT GIS and LBNL Queued Up datasets, detected **210 material queue events**, and generated a **2,206-row entity-resolution audit** with source traces, confidence signals, file hashes, mismatches, and limitations.
- Designed deterministic high-stakes agent evaluations for unsupported claims, missing citations, weak-sample handling, ambiguous entity matches, and overclaiming of pending rules; deployed a public browser-based demo for end-to-end testing.

Awards

Kaggle Competitions: Silver Medal — OTTO Recommender System; Silver Medal — G2Net Gravitational Waves; Bronze Medal — RSNA Cancer Detection